LAB # 7
To Reproduce a Network and Show Successful Connectivity between Hosts where OSPF is Configured in Routers using CISCO Packet Tracer

Objectives

- To construct a network for demonstration of the operation of OSPF routing protocol using Cisco Packet Tracer
- To show the connectivity between nodes in a network using OSPF on all routers.

Pre-Lab Exercise

Read this experiment in its entirety to become familiar with objectives of this lab. Study in detail and become familiar with the Link State Routing basics provided with this laboratory experiment and in the chapter 4 of the reference book. You may record the terms and sections that require more elaboration for reference. The instructor may provide the class some time to reflect upon these before proceeding with the lab.

Open Shortest Path First (OSPF) Basics

Open Shortest Path First (OSPF) is an open standard link state routing protocol that's been implemented by a wide variety of network vendors, including Cisco. And it's that open standard characteristic that's the key to OSPF's flexibility and popularity.

OSPF works by using the Dijkstra algorithm to initially construct a shortest path tree and follows that by populating the routing table with the resulting best paths. It is quickly convergent. Another two great advantages OSPF offers are that it supports multiple, equal-cost routes to the same destination, also supports both IP and IPv6 routed protocols. OSPF's best features are:

- Allows for the creation of areas and autonomous systems
- Minimizes routing update traffic
- Is highly flexible, versatile, and scalable
- Supports VLSM/CIDR
- Offers an unlimited hop count
- Is open standard and supports multi-vendor deployment

Here are three of the biggest reasons to implement OSPF in a way that makes full use of its intentional, hierarchical design:

- To decrease routing overhead
- To speed up convergence
- To confine network instability to single areas of the network

OSPF Terminology

Imagine being given a map and compass with no prior concept of east, west, north or south—not even what rivers, mountains, lakes, or deserts are. I'm guessing that without any ability to orient yourself in a basic way, your cool, new tools wouldn't help you get anywhere but completely lost, right? This is exactly why we're going to begin exploring OSPF by getting you solidly acquainted with a fairly long list of terms before setting out from base camp into the great unknown! Here are those vital terms to commit to memory now:

OSPF Metrics

OSPF uses a metric referred to as *cost*. A cost is associated with every outgoing interface included in an SPF tree. The cost of the entire path is the sum of the costs of the outgoing interfaces along the path. Cisco uses a

simple equation of 10^8/*bandwidth*, where *bandwidth* is the configured bandwidth for the interface. Using this rule, a 100 Mbps Fast Ethernet interface would have a default OSPF cost of 1.

Link

A *link* is a network or router interface assigned to any given network. When an interface is added to the OSPF process, it's considered to be a link. This link, or interface, will have up or down state information associated with it as well as one or more IP addresses.

Router ID

The *router ID (RID)* is an IP address used to identify the router. Cisco chooses the router ID by using the highest IP address of all configured loopback interfaces. If no loopback interfaces are configured with addresses, OSPF will choose the highest IP address out of all active physical interfaces. To OSPF, this is basically the "name" of each router.

Neighbor

Neighbors are two or more routers that have an interface on a common network, such as two routers connected on a point-to-point serial link. OSPF neighbors must have a number of common configuration options to be able to successfully establish a neighbor relationship, and all of these options must be configured exactly the same way:

- Area ID
- Stub area flag
- Authentication password (if using one)
- Hello and Dead intervals

Adjacency

An *adjacency* is a relationship between two OSPF routers that permits the direct exchange of route updates. Unlike EIGRP, which directly shares routes with all of its neighbors, OSPF is really picky about sharing routing information and will directly share routes only with neighbors that have also established adjacencies. And not all neighbors will become adjacent—this depends upon both the type of network and the configuration of the routers. In multi-access networks, routers form adjacencies with designated and backup designated routers. In point-to-point and point-to-multipoint networks, routers form adjacencies with the router on the opposite side of the connection.

Designated Router

A *designated router (DR)* is elected whenever OSPF routers are connected to the same broadcast network to minimize the number of adjacencies formed and to publicize received routing information to and from the remaining routers on the broadcast network or link. Elections are won based upon a router's priority level, with the one having the highest priority becoming the winner. If there's a tie, the router ID will be used to break it. All routers on the shared network will establish adjacencies with the DR and the BDR which ensures that all router's topology tables are synchronized.

Backup Designated Router

A *backup designated router (BDR)* is a hot standby for the DR on broadcast, or multi-access, links. The BDR receives all routing updates from OSPF adjacent routers but does not disperse LSA updates.

Hello protocol The OSPF Hello protocol provides dynamic neighbor discovery and maintains neighbor relationships. Hello packets and Link State Advertisements (LSAs) build and maintain the topological database. Hello packets are addressed to multicast address 224.0.0.5.

Neighborship Database

The *neighborship database* is a list of all OSPF routers for which Hello packets have been seen. A variety of details, including the router ID and state, are maintained on each router in the neighborship database.

Topological Database

The *topological database* contains information from all of the Link State Advertisement packets that have been received for an area. The router uses the information from the topology database as input into the Dijkstra algorithm that computes the shortest path to every network.

Link State Advertisement

A *Link State Advertisement (LSA)* is an OSPF data packet containing link-state and routing information that's shared among OSPF routers. An OSPF router will exchange LSA packets only with routers to which it has established adjacencies.

OSPF Areas

An *OSPF area* is a grouping of contiguous networks and routers. All routers in the same area share a common area ID. Because a router can be a member of more than one area at a time, the area ID is associated with specific interfaces on the router. This would allow some interfaces to belong to area 1 while the remaining interfaces can belong to area 0. All of the routers within the same area have the same topology table. When configuring OSPF with multiple areas, you've got to remember that there must be an area 0 and that this is typically considered the backbone area. Areas also play a role in establishing a hierarchical network organization—something that really enhances the scalability of OSPF!

Broadcast (multi-access)

Broadcast (multi-access) networks such as Ethernet allow multiple devices to connect to or access the same network, enabling a *broadcast* ability in which a single packet is delivered to all nodes on the network. In OSPF, a DR and BDR must be elected for each broadcast multi-access network.

Non-Broadcast multi-access

Non-Broadcast multi-access (NBMA) networks are networks such as Frame Relay, X.25, and Asynchronous Transfer Mode (ATM). These types of networks allow for multi-access without broadcast ability like Ethernet. NBMA networks require special OSPF configuration to function properly.

Point-to-Point

Point-to-point refers to a type of network topology made up of a direct connection between two routers that provides a single communication path. The point-to-point connection can be physical—for example, a serial cable that directly connects two routers—or logical, where two routers thousands of miles apart are connected by a circuit in a Frame Relay network. Either way, point-to-point configurations eliminate the need for DRs or BDRs.

Point-to-Multipoint

Point-to-multipoint refers to a type of network topology made up of a series of connections between a single interface on one router and multiple destination routers. All interfaces on all routers share the point-to-multipoint connection and belong to the same network. Point-to-multipoint networks can be further classified according to whether they support broadcasts or not. This is important because it defines the kind of OSPF configurations you can deploy.

In-Lab Exercise

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Figure 14. Assign Network_ID from Class B i.e. 172.17.0.0/24 and fill in the table. The segments of the network have been subnetted using VLSM.OSPF is a classless routing protocol that can be used to provide subnet mask information in the routing updates. This will allow VLSM subnet information to be propagated throughout the network. You will then perform the initial router configurations required for connectivity. Use the IP addresses according to your IP assignment. First test the connections between directly connected devices, and then test connectivity between devices that are not directly connected. You will configure the dynamic routing protocol OSPF to add remote networks, this will allow communication between the hosts of different networks. View the routing table after each dynamic route is added to observe how the routing table has changed.

Topology Diagram

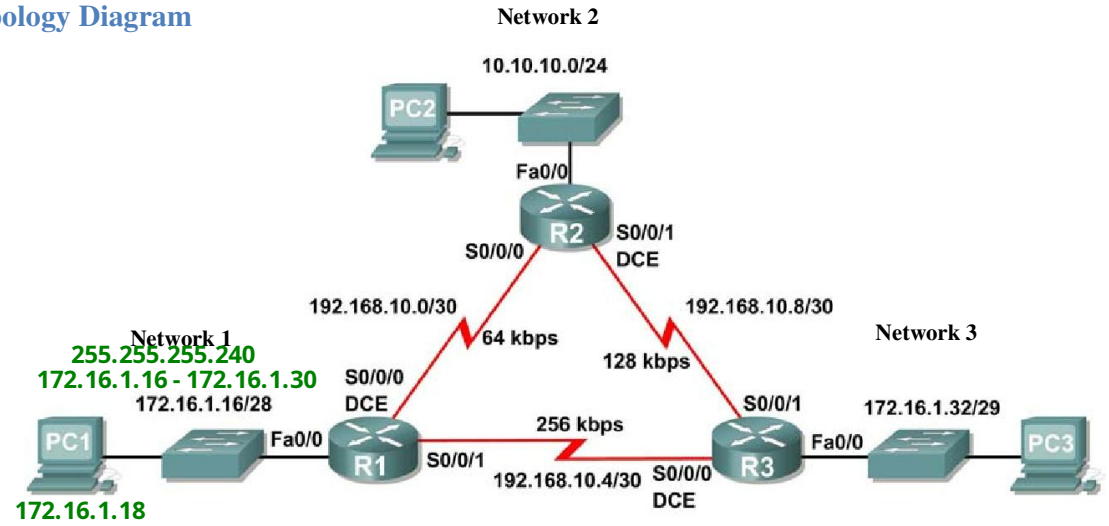

Figure 14: Network Topology

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	172.16.1.17	255.255.255.240	
	S0/0/0	192.168.10.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	Fa0/0	10.10.10.1	255.255.255.0	
	S0/0/0	192.162.10.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
R3	Fa0/0	172.16.1.33	255.255.255.248	
	S0/0/0	192.168.10.6	255.255.255.252	
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.18	255.255.255.240	172.16.1.17
PC2	NIC	10.10.10.2	255.255.255.0	10.10.10.1

PC3	NIC	172.16.1.34	255.255.255.248	172.16.1.33

Task 1: Prepare the Network

Task 2: Perform Basic Router Configurations.

Step 1: Configure and Activate Serial and Ethernet interfaces on R1, R2, and R3.

Step 2: Verify IP addressing and interfaces.

Use the *show ip interface brief* command to verify that the IP addressing is correct and that the interfaces are active.

Task 3: Perform IP configuration on host PC's

Task 4: Configure OSPF

Step 1: Learn how to enable OSPF

Router(config)#**router ospf***Process ID*

Router(config-router)#**network** *Network1 Wildcard mask* **area** *OSPF areas*

Router(config-router)#**network***Network2 Wildcard mask* **area** *OSPF areas*

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Router(config-router)#**network***NetworkN Wildcard mask* **area** *OSPF areas*

Process ID: A value in the range from 1 to 65,535 identifies the OSPF process ID. It's a unique number on this router that groups a series of OSPF configuration commands under a specific running process. Different OSPF routers don't have to use the same process ID to communicate. It's a purely local value that doesn't mean a lot, but you still need to remember that it cannot start at 0; it has to start at a minimum of 1.

Network – Network that is directly connected with that router. You have to enter each directly connected network (range from 1 to N) separately using ***network*** command

Wildcard mask – Inverse of a subnet mask. For Example, the inverse of the subnet mask255.255.255.0 is 0.0.0.255. To calculate the inverse of the subnet mask, subtract the subnet mask from 255.255.255.255:

255.255.255.255

– 255.255.255.0 Subtract the subnet mask

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0. 0. 0. 255 Wildcard mask

OSPF areas–An *OSPF area* is a grouping of contiguous networks and routers. All routers in the same area share a common area ID. Because a router can be a member of more than one area at a time, the area ID is associated with specific interfaces on the router. This would allow some interfaces to belong to area 1 while the remaining interfaces can belong to area 0. All of the routers within the same area have the same topology table. When configuring OSPF with multiple areas, you've got to remember that there must be an area 0 and that this is typically considered the backbone area. Areas also play a role in establishing a hierarchical network organization—something that really enhances the scalability of OSPF!

Task 5: Configure OSPF on the R1 Router

Step 1: Enable OSPF. Use the *router ospf* command in global configuration mode to enable OSPF on the R1 router. Enter a process ID of 1 for the *process-ID* parameter.

R1(config)#**router ospf 1**

Step 2: Configure the networks

Once you are in the Router OSPF configuration sub-mode, configure the networks to be included in the OSPF updates that are sent out of R1 i.e. networks directly connected with R1.

Step 3: Configure the Network 1

Wildcard mask has been calculated for this network in wildcard mask example i.e. 0.0.0.255. Complete the following command and configure it on Router 1. Enter OSPF area of 0 for the *OSPF areas* parameter. You will use same OSPF area for whole lab.

R1(config-router)#**network** network 172.16.1.16 0.0.0.15 area 0

Step 4: Configure the router to advertise the networks attached to the Serial0/0/0 and Serial0/0/1 interface respectively.

Calculate the wildcard mask for the network, complete the following commands and enter in Router OSPF configuration sub-mode.

R1(config-router)#**network** 192.168.10.0 0.0.0.3 **area** *0*

R1(config-router)#**network** 192.168.10.4 0.0.0.3 **area** *0*

Step 5: When you are finished with the OSPF configuration for R1, return to privileged EXEC mode and save the current configuration to NVRAM.

R1(config-router)#**end**

Task 6: Configure OSPF on the R2 and R3 Routers

Step 1: Enable OSPF routing on the R2 router. Write the command to enable OSPF.

router ospf 1

Step 2: Configure the router R2 to advertise the directly connected networks. Write down the commands and run in Router OSPF configuration sub-mode.

R2(config-router)# network 10.10.10.0 0.0.0.255 area 0
R2(config-router)# network 192.168.10.0 0.0.0.3 area 0
R2(config-router)# network 192.168.10.8 0.0.0.3 area 0
R2(config-router)# end

Step 3: Notice that Link State protocol sends a notification message to the console stating that a neighbor relationship with another OSPF router has been established.

What is the IP address of the OSPF neighbor router?

192.168.10.1

What interface on the R2 router is the neighbor adjacent to?

_____

Step 4: Enable OSPF routing on the R3 router. Write the command to enable OSPF.

_____

```
router ospf 1
```

_____

Step 5: Configure the router R3 to advertise the directly connected networks. Write down the commands and run in Router OSPF configuration sub-mode.

_____

```
R3(config-router)# network 172.16.1.32 0.0.0.7 area 0
R3(config-router)# network 192.168.10.4 0.0.0.3 area 0
R3(config-router)# network 192.168.10.8 0.0.0.3 area 0
R3(config-router)# end
```

_____

Task 7: Configure OSPF Router IDs

The OSPF router ID is used to uniquely identify the router in the OSPF routing domain. A router ID is an IP address. Cisco routers derive the Router ID in one of three ways and with the following precedence:

- IP address configured with the OSPF *router-id* command.
- Highest IP address of any of the router's loopback addresses.
- Highest active IP address on any of the router's physical interfaces.

Step 1: Examine the current router IDs in the topology.

Since no router IDs or loopback interfaces have been configured on the three routers, the router ID for each router is determined by the highest IP address of any active interface. Run the following command on each router to check router ID

Router#**showip protocols**

What is the router ID for R1?

192.168.10.5

_____

What is the router ID for R2?

192.168.10.9

_____

What is the router ID for R3?

192.168.10.10

_____

Step 2: Use loopback addresses to change the router IDs of the routers in the topology.

R1(config)#**interface loopback** 0

R1(config-if)#**ip address** 10.1.1.1 255.255.255.255

R2(config)#**interface loopback** 0

R2(config-if)#**ip address** 10.2.2.2 255.255.255.255

R3(config)#**interface loopback** 0

R3(config-if)#**ip address** 10.3.3.3 255.255.255.255

Step 3: Reload the routers to force the new Router IDs to be used.

When a new Router ID is configured, it will not be used until the OSPF process is restarted. Make sure that the current configuration is saved to NRAM, and then use the **reload** command to restart each of the routers.

When the router is reloaded, what is the router ID for R1? 10.1.1.1

When the router is reloaded, what is the router ID for R2? 10.2.2.2

When the router is reloaded, what is the router ID for R3? 10.3.3.3

Step 4: Use the show *ipospfneighbor*s command to verify that the router IDs have changed.

Task 8: Verify OSPF Operation

Step 1: On the R1 router, Use the *show ipospfneighbor* command and write your observations.

R1#**show ipospfneighbor**

```
r1 is connected to 2 neighbours having ids:
    10.2.2.2 -> deadtime 34s . state full
    10.3.3.3 -> deadtime 35s . state full
```

Step 2: On the R1 router, use the *show ip protocols* command and write your observations.

R1#**show ip protocols**

```
routing protocol is ospf 1 , router id is 10.1.1.1

shows OSPF neighbors

Distance: 110
```

Step 3: Examine OSPF Routes in the Routing Tables

R1#**show ip route**

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.1.1.1/32 is directly connected, Loopback0
O    10.10.10.0/24 [110/455] via 192.168.10.6, 00:19:55, Serial3/0
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.1.16/28 is directly connected, FastEthernet0/0
O    172.16.1.32/29 [110/391] via 192.168.10.6, 00:19:55, Serial3/0
192.168.10.0/30 is subnetted, 3 subnets
C    192.168.10.0 is directly connected, Serial2/0
C    192.168.10.4 is directly connected, Serial3/0
O    192.168.10.8 [110/454] via 192.168.10.6, 00:19:55, Serial3/0
```

Step 4: Repeat Step 1 to Step 3 for Router R2 and R3

On R2:

```
R2# show ip ospf neighbor
R2# show ip protocols
R2# show ip route
```

n R3:

R3# show ip ospf neighbor
R3# show ip protocols
R3# show ip route

Step 5: Use ping command to access PC1 to PC2 and PC1 to PC3 verify that OSPF is configured correctly. Is ping successful?

yes

Step 6: Use the show interfaces serial0/0/0 command on the R1 router to view the bandwidth of the Serial 0/0/0 interface. What is the default bandwidth of link?

R1#show interfaces serial0/0/0

BW 64 Kbit

Step 7: Calculate the cost of link to find metrics value using the formula 10^8/bandwidth. Is calculate and router measured value same?

cost ==> 100 000 000 / 1544 000= 64.77, routers measure is 64

Task 9: Configure Additional Features

Step 1: Use the bandwidth command to change the bandwidth of the serial interface serial0/0/0 of the R1 and R2 routers to the actual bandwidth, 64 kbps.

R1 router:

R1(config)#**interface serial0/0/0**

R1(config-if)#**bandwidth 64**

R2 router:

R2(config)#**interface serial0/0/0**

R2(config-if)#**bandwidth 64**

Step 2: Use the *show ipospf interface* command on the R1 router to verify the cost of the serial links. What is new cost of serial0/0/0 link?

100,000,000 / 64,000 = 1562.5 , router measure is 1562

Step 3: Use the *show ipospfneighbor* command on R1 to view the Dead Time counter. What is default dead interval and what do you observe?

_____

dead time : 33

_____

Step 4: Configure the OSPF Hello and Dead intervals.

The OSPF Hello and Dead intervals can be modified manually using the*ipospf hello-interval*and*ipospf dead-interval*interface commands. Use these commands to changethe hello interval to 5 seconds and the dead interval to 20 seconds on the Serial 0/0/0 interface of the R1 router.

R1(config)#**interface serial0/0/0**

R1(config-if)#**ipospf hello-interval 5**

R1(config-if)#**ipospf dead-interval 20**

Step 5: Modify the Dead Timer and Hello Timer intervals on the Serial 0/0/0 interface in the R2 router to match the intervals configured on the Serial 0/0/0 interface of the R1 router.

R2(config)#**interface serial0/0/0**

R2(config-if)#**ipospf hello-interval 5**

R2(config-if)#**ipospf dead-interval 20**